DIPLOMA PROGRAMME PRACTICE · SL · SET B

Biology / Paper 1A

Question booklet · 30 marks · 90 minutes across 1A and 1B

Answer every question. Choose one option, A–D, per question. Use a calculator. Record choices online or on separate paper.

QUESTION 1 [1]

A leaf cools as liquid water evaporates. Which molecular event accounts for the energy transferred from the leaf?

A Covalent O–H bonds in liquid molecules must be broken before evaporation
B Energy released as hydrogen bonds form passes directly from vapour back to the leaf
C Overcoming intermolecular attractions allows higher-energy water molecules to escape
D Only the least energetic water molecules enter the vapour, cooling the liquid

QUESTION 2 [1]

A DNA strand has sequence 5′-AGTCCA-3′. Which sequence is the complementary strand written in the 5′ to 3′ direction?

A 5′-TCAGGT-3′
B 5′-UGGACU-3′
C 5′-AGTCCA-3′
D 5′-TGGACT-3′

QUESTION 3 [1]

Unlabelled cells contain ribosomes and DNA. Which additional observation provides the clearest evidence of eukaryotic organization?

A A membrane-bounded nucleus encloses the chromosomes
B The cells synthesize polypeptides
C The cells divide
D The cells have a plasma membrane

QUESTION 4 [1]

A field worker identifies a specimen as Panthera leo. Which inference follows from the binomial alone?

A Its population is reproductively isolated from every other population of lions
B It shares its genus with Panthera pardus
C It belongs to the same species as every Panthera
D Its closest living relative must be Panthera pardus

QUESTION 5 [1]

Two insect populations are separated by a river. Which process directly reduces gene flow and permits independent evolutionary change?

A The barrier reduces successful movement and mating between the populations
B Phenotypic differences automatically arise only after hybrid sterility evolves
C Equal mutation rates make reproductive isolation inevitable within one generation
D Both groups show the same environmental modification without a heritable difference

QUESTION 6 [1]

A habitat contains 12 butterfly species before clearance and 12 after clearance. Which additional result shows a loss of biodiversity that richness alone misses?

A Both counts are made in summer
B Several formerly common native species are replaced by widespread generalists
C Sampling effort is identical
D Butterflies were sampled at noon on both dates

QUESTION 7 [1]

Two triacylglycerols have equal chain lengths. One has more cis double bonds and remains liquid at a lower temperature. Why?

A Double bonds cause permanent positive charges on every chain
B Unsaturated lipids contain no covalent bonds
C Cis double bonds bend chains and reduce close packing
D Cis double bonds shorten the glycerol molecule

QUESTION 8 [1]

A polypeptide contains 96 amino acid residues in one unbranched chain. How many water molecules were released when free amino acids formed its peptide bonds?

A 96
B 192
C 95
D 97

QUESTION 9 [1]

An artificial bilayer encloses a solution of KCl. Which substance crosses its lipid core most readily in the absence of proteins?

A K^+
B A large soluble protein
C O_2
D Cl^-

QUESTION 10 [1]

Lysosomal enzymes work best at a lower pH than enzymes in the cytosol. What advantage follows from a lysosomal membrane?

A It prevents any transfer of products from the lysosome to the cytosol
B It keeps substrates permanently outside while soluble enzymes digest them from within
C It raises the lysosomal pH to match the cytosol during normal digestion
D It maintains a separate chemical environment for digestion

QUESTION 11 [1]

An intestinal epithelial cell has many microvilli on its lumen-facing membrane. Which consequence follows most directly?

A More surface is available for transport proteins per cell
B The number of chromosomes increases
C All solutes can enter without transport proteins
D The diffusion distance across each phospholipid bilayer doubles

QUESTION 12 [1]

In a sealed bell-jar model of ventilation, pulling down the flexible membrane expands the balloons. What immediately drives air into them?

A Carbon dioxide actively pumps air through the tube
B The pressure outside the jar suddenly falls
C Oxygen diffusion causes the model diaphragm to contract
D Pressure inside the balloons falls below external pressure

QUESTION 13 [1]

A student removes a ring of bark containing phloem from a stem while leaving the xylem intact. Sugars accumulate above the ring. Which inference is best supported?

A Transport of assimilates from leaves to sinks below the ring is interrupted
B The leaves can no longer absorb light
C Xylem normally transports all sugars downward
D The root immediately stops receiving all water

QUESTION 14 [1]

Two aquatic mammals have similar shapes. Species X has twice the linear dimensions of Y. Under identical conditions, which prediction follows from geometry alone?

A X has twice the surface-area-to-volume ratio
B X loses less heat per unit body volume
C X generates heat only at its skin
D X has one quarter the body volume

QUESTION 15 [1]

In a feeding experiment one insect drinks nectar and carries pollen between flowers of the same plant species. Which result establishes mutualism most directly?

A Access increases both insect reproductive output and viable seed production relative to suitable controls

B The insect removes pollen and has higher reproductive output, while seed production is unchanged

C Flowers receiving more visits produce less viable seed, but insects collect more nectar

D Flowers set more seed when visited, while insect reproductive output declines

QUESTION 16 [1]

An enzyme assay compares temperatures using identical concentrations. At 55 °C the initial rate is high but activity soon falls even after transfer back to 25 °C with fresh substrate. What best explains the persistent decline?

A Every enzyme molecule gained another active site

B Cooling permanently increases collision frequency

C Heat altered the functional conformation of some enzyme molecules

D The original substrate was consumed and the fresh substrate cannot react

QUESTION 17 [1]

Equal numbers of yeast cells supplied with glucose release CO_2 both with and without oxygen. Which conclusion follows?

A CO_2 release alone cannot distinguish aerobic respiration from alcoholic fermentation

B Both treatments yield equal ATP per glucose

C The oxygen-free culture cannot regenerate NAD^+

D CO_2 production proves that oxygen was consumed

QUESTION 18 [1]

Unvariegated leaves are exposed to light of different colours at equal photon flux. Which measurement is the most direct evidence of a difference in net photosynthetic rate?

A Wavelength of light reflected from dead tissue

B CO_2 uptake per unit leaf area per minute

C Final leaf area alone

D Total chlorophyll mass before illumination

QUESTION 19 [1]

At a typical chemical synapse, why does information normally pass from the presynaptic neuron to the postsynaptic cell?

A Both cells release identical transmitter simultaneously

B Action potentials cross the cleft as a continuous flow of cytoplasm

C Transmitter release machinery and postsynaptic receptors are arranged asymmetrically

D Neurotransmitters can diffuse in only one direction in water

QUESTION 20 [1]

A small rise in body temperature is followed by increased sweating. Which observation completes a negative feedback loop?

A Sweat rate remains unchanged when the initial rise in temperature is reversed
B Cooling activates a further increase in sweating that drives temperature away from its set range
C Sweating increases in direct proportion to exercise regardless of sensed temperature
D As temperature approaches its set range, stimulation of sweating declines

QUESTION 21 [1]

After an effective antibiotic is used repeatedly, resistant bacteria form a larger proportion of a population. Which explanation is best supported?

A Human antibodies permanently change bacterial DNA
B Susceptible cells leave fewer descendants than cells carrying resistance
C The antibiotic directs the same useful mutation in every cell
D Each bacterium intentionally mutates when exposed

QUESTION 22 [1]

Ten randomly located quadrats, each 0.25 m², contain 35 seedlings in total. At the estimated mean density, how many seedlings occupy 20 m²?

A 70
B 280
C 700
D 175

QUESTION 23 [1]

A tracer enters the carbon atoms of grass via photosynthesis. Which route can return the same carbon to the atmosphere without combustion?

A Grass → grazer → respiratory CO_2
B Grass → ATP phosphate → atmospheric oxygen
C Grass → rainfall → atmospheric nitrogen
D Grass → sunlight → atmospheric O_2

QUESTION 24 [1]

A replication system receives all four deoxyribonucleotides but lacks a template. Which property of normal DNA replication is directly lost?

A Incoming nucleotides lose their complementary hydrogen-bonding groups
B The number of types of deoxyribonucleotide available falls from four to two
C The direction of movement of DNA polymerase along its template reverses
D Complementary pairing that specifies the new strand sequence

QUESTION 25 [1]

An mRNA region contains 75 nucleotides from the first base of the start codon through the last base of the stop codon, inclusive. How many amino acids are encoded?

A 25
B 24
C 72
D 26

QUESTION 26 [1]

A single base is inserted early in a coding region. Which consequence is most likely if no second mutation compensates?

A One whole amino acid is always added without affecting later codons
B The DNA molecule contains no paired bases after insertion
C Every amino acid before the insertion changes
D The grouping into downstream codons shifts

QUESTION 27 [1]

At which event of meiosis is chromosome number per daughter nucleus reduced from diploid to haploid?

A Pairing of bases during interphase
B Separation of homologous chromosomes in meiosis I
C Separation of sister chromatids in mitosis
D DNA replication before meiosis I

QUESTION 28 [1]

A seed contains an embryo with alleles from two parents. Which event created the diploid cell from which this embryo developed?

A Pollen transfer to a stigma
B Dispersal of the mature seed
C Fusion of male and female gamete nuclei
D Formation of haploid pollen by meiosis

QUESTION 29 [1]

An individual secretes too little insulin after a meal. Which response is most directly impaired?

A Return of elevated blood glucose toward its usual range
B The immediate rise in glucose caused by intestinal absorption
C Glycogen breakdown in the liver in response to glucagon
D Secretion of glucagon by pancreatic alpha cells during fasting

QUESTION 30 **[1]**

Two years have the same atmospheric CO_2 concentration but different global mean temperatures. Which conclusion is justified?

A A single year establishes a long-term cooling trend

B Every climate variable is determined by CO_2 concentration alone

C CO_2 absorbs no infrared radiation

D Short-term variability can coexist with a long-term greenhouse effect